

Chapter Four Advanced Techniques (The Heart of Prompt Engineering)

4-1: Chain of Thought (CoT) — Forcing the Model to Show Its Reasoning Path

What's the story?

AI is inherently a “fast-response machine.” If you ask “*What is 27 times 56?*”, it directly guesses the answer from memory or repetitive patterns (and may be wrong). But the **CoT** technique tells it: “Don’t rush! Go step by step and write each stage for me until you reach the answer.”

Breaking the model's brain open like this:

- **Without CoT:** “*Write a weekly plan for weight loss.*” → Response: a general plan.
- **With CoT (chained):** “*I want you to write a weekly plan for weight loss. Please **think step by step and stage by stage** and write what I should calculate first? (target weight, basal calories), then what meals come next in the arrangement? And finally, how do I monitor that I'm on track? Explain each step separately.*”

Golden tip: The phrase “**Let's think step by step**” is a magic trick. According to research, this one simple phrase raises the accuracy of mathematical and logical answers from 30% to above 70%. When the model is forced to write its reasoning, it corrects its own intermediate mistakes.

4-2: Tree of Thoughts (ToT) — Exploring Multiple Solutions Simultaneously

If CoT is a **straight line** (step one → two → three), ToT is a **garden tree**. Here we tell the model: “Don’t take one path; rather, at each stage, examine 3 or 4 different solutions together, compare them with each other, and finally choose the best one.”

This technique works wonders for creative, strategic, and puzzle-like problems.

Practical example:

*“I want to choose an advertising slogan for my clothing brand. Instead of one slogan, **explore 3 different paths** in parallel:*

- *Path 1: Classic and enduring slogans (like Nike).*
- *Path 2: Humorous and youth-friendly slogans (like Snap).*

- *Path 3: Emotional and story-driven slogans (like Digikala).*

Explain each path with 2 suggested slogan examples, then at the end tell me which path you think is more suitable for a sportswear brand and why?”

See? We no longer have one answer; we have an **idea-generation workshop** that cherry-picks the best option.

4-3: Self-Consistency — Ask the Model to Answer Multiple Times and Choose the Best

Here we turn a weakness of the model into a strength. We know that AI is unpredictable and if you ask a question 3 times, it may give 3 different answers (due to the same “probability” technique we mentioned in Chapter One).

The **Self-Consistency** technique means: **“Answer this question 3 times, then review all 3 answers yourself and tell me which one is more correct, more comprehensive, and more logical, and why you reject the other two.”**

This causes:

1. The model’s random errors to be eliminated (because if one answer is wrong, the other two veto it).
2. The most comprehensive answer to be selected (because the model sees the strengths of all three).

Example:

*“Solve the following logical problem: ‘If all cats are carnivores and some pets are carnivores, do we conclude that some pets are cats?’ Please solve this problem **3 times** with complete reasoning (each time independent of the previous). Then review all 3 answers yourself and tell me which reasoning is the strongest and without logical contradiction.”*

4-4: Conditional Prompts (If X, Go to Y, Otherwise Z) — If-Else Logic

Here we are talking to AI like a programmer. A conditional prompt tells the model: “Your behavior depends on my input. If such-and-such happens, answer differently.”

This technique is very critical for building chatbots, support assistants, and decision-making systems.

How to write it?

Very simply, use words like “if... otherwise...” and “in case... otherwise...”

Comprehensive example:

“You are a technical support assistant. Act as follows:

- If the user says **they forgot their password**, give the password recovery link and explain the steps step by step.*
- If the user says **they have a payment issue**, ask them what the amount and destination bank are and request the tracking number.*
- If the user says **hello** or greets you, just give a polite response and ask what you can help with (and do not enter any of the technical explanations).*

Now I say: ‘Hi, I couldn’t make a payment.’ — Respond exactly according to the conditions above!”

The model checks: “Hi” means the third condition, but “I couldn’t make a payment” means the second condition. Since the second condition is more specific, it gives the response related to the payment issue.

Quick Comparison Table (Which Technique Should I Use?)

Technique	Best Use Case	Complexity
CoT (Chained)	Math, coding, root-cause analysis	Medium
ToT (Tree)	Ideation, strategy, choosing between options	High
Self-Consistency	Precise and error-sensitive questions (medical, legal, engineering)	High
Conditional (If-Else)	Building support bots, smart forms, interactive guides	Medium

Combined Practical Project for Chapter Four (So You Fully Master It!)

Now that you’ve learned these 4, I’ll write a prompt that contains **all 4 techniques**. See how it turns a simple task into a masterpiece:

“As an investment advisor [Role], I have 100 million tomans and I want to invest in 3 areas (gold, housing, stocks) [Context].

[CoT] Please write the decision-making steps for each area step by step for me.

[ToT] Then for the stocks area, examine 3 different sub-strategies (short-term, long-term,

mixed) in parallel and write the advantages of each.

[Self-Consistency] *Now calculate the best capital allocation ratio among these 3 areas 2 more times (3 times total) and finally give me the final consensus from among these 3 calculations.*

[Conditional] *In your responses, if inflation is above 30%, lean your suggestion toward gold; otherwise, lean your suggestion toward housing. (Assume inflation is now 35%.) Deliver the output in the form of a table with 4 columns (area, amount of capital, reason, risk)."*

Important practical assignment for Chapter Four (Send It So I Can See How Professional You've Become):

Suppose you want to guide an AI for **"diagnosing a car problem."** Write a prompt in which:

1. You use the **CoT** technique (it explains the troubleshooting steps step by step).
2. You set a condition (If-Else) that if the sound comes from the engine, it takes one path, and if the sound comes from the brakes, it takes another path.